

STEAMBOAT SPRINGS, CO, USA

WMO: 724678

Lat: 40.517N

Lon: 106.867W

Elev: 6879

StdP: 11.39

Time Zone: -7.00 (NAM)

Period: 11-19

WBAN: 94033

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-1.4	-12.7	3.6	1.4	-7.1	4.8	8.2	18.5	27.0	16.0	26.8	0.3	100	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.8	87.0	55.0	84.8	54.8	82.5	54.4	60.0	76.2	58.7	75.5	57.4	74.1	7.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.8	85.8	61.2	53.5	78.8	61.5	51.5	73.1	61.2	29.8	76.8	28.8	75.1	27.7	74.1	66.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.1	12.5	DB	-17.3	89.9	6.5	3.1	-22.0	92.1	-25.9	93.9	-29.5	95.7	-34.3	98.0	(4)
(5)			WB	-17.8	61.0	6.4	1.2	-22.4	61.9	-26.2	62.6	-29.7	63.3	-34.4	64.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.2	18.4	23.9	33.0	40.6	48.8	60.1	66.1	63.7	56.7	43.1	31.5	19.1	
	DBStd	17.98	9.42	8.38	7.99	6.66	6.50	5.56	3.43	3.57	5.74	8.47	8.40	9.50	
	HDD50	4390	980	730	528	287	100	5	0	0	14	234	555	957	
	HDD65	8437	1445	1150	992	731	501	162	29	67	253	680	1005	1422	
	CDD50	1531	0	0	1	6	63	307	498	423	215	18	0	0	
	CDD65	108	0	0	0	0	0	15	63	25	5	0	0	0	
	CDH74	3832	0	0	0	1	33	866	1530	1067	335	0	0	0	
	CDH80	1069	0	0	0	0	1	237	490	283	58	0	0	0	
Wind	WSAvg	3.1	1.4	2.4	2.8	4.4	4.5	3.9	3.5	3.5	3.4	3.0	2.2	1.7	
Precipitation	PrecAvg	22.6	2.4	2.0	1.9	2.2	2.0	1.5	1.4	1.6	1.9	1.7	2.1	2.3	
	PrecMax	35.5	5.9	4.7	3.2	4.1	5.7	5.6	3.3	3.5	7.6	4.0	5.6	6.3	
	PrecMin	16.0	0.6	0.8	0.6	0.6	0.3	0.1	0.1	0.4	0.2	0.1	0.5	0.6	
	PrecStd	4.5	1.3	0.9	0.7	0.9	1.1	1.1	0.8	0.8	1.5	1.0	1.1	1.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.8	47.8	64.2	70.8	78.1	89.7	90.2	88.1	85.9	72.4	62.0	44.2	
		MCWB	32.4	36.7	41.8	43.5	48.0	53.6	54.2	54.3	54.9	48.3	41.9	34.0	
	2%	DB	37.8	43.0	57.6	65.6	73.9	85.9	87.3	85.4	81.3	68.7	55.7	39.3	
		MCWB	31.3	33.7	39.7	43.5	47.4	53.5	55.7	54.4	52.3	46.4	39.7	31.9	
	5%	DB	34.4	39.7	52.2	61.2	70.5	82.4	85.3	83.5	77.7	65.3	51.0	35.1	
		MCWB	30.0	32.8	37.7	41.8	46.7	53.2	55.7	54.2	51.0	45.1	37.6	29.7	
	10%	DB	31.9	36.7	47.4	56.2	66.6	79.5	82.9	80.9	74.9	60.8	46.1	32.2	
		MCWB	28.9	31.5	35.5	40.3	46.3	52.1	55.6	54.2	50.7	43.7	35.6	28.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.4	37.8	42.9	46.4	51.7	59.2	62.5	61.2	56.9	52.2	43.5	36.0
			MCDB	37.6	44.7	61.7	64.7	70.4	80.9	75.8	77.0	76.6	65.7	57.0	43.5
2%		WB	32.8	34.9	40.7	44.5	50.0	57.3	60.4	59.5	55.4	49.5	40.9	33.0	
		MCDB	35.9	40.1	56.0	61.6	68.1	77.6	75.7	75.8	72.3	63.9	53.6	37.2	
5%		WB	31.1	33.4	38.2	43.1	48.4	55.7	59.4	58.0	54.2	46.9	38.7	30.9	
		MCDB	33.6	38.1	50.3	57.7	65.6	75.4	75.8	73.3	70.3	60.8	49.6	34.4	
10%		WB	28.8	32.1	36.1	41.2	47.1	54.2	58.3	56.4	52.9	44.7	36.3	28.8	
		MCDB	31.8	36.4	45.8	54.1	63.4	74.1	74.7	72.2	68.2	58.5	44.9	31.8	
Mean Daily Temperature Range	5% DB	MDBR	19.5	19.5	23.0	23.2	26.6	34.7	32.8	33.4	32.2	26.8	22.9	19.0	
		MCDBR	21.5	21.4	29.5	33.4	35.4	39.8	37.1	38.3	38.4	36.2	30.5	22.0	
		MCWBR	15.3	13.4	14.3	14.5	13.8	14.5	11.9	12.8	14.6	16.6	16.1	15.2	
	5% WB	MCDBR	16.4	17.8	27.1	29.0	30.9	35.2	31.0	30.9	30.9	30.2	27.3	18.2	
		MCWBR	12.6	11.0	13.4	13.4	13.2	14.3	11.7	11.7	12.0	14.6	14.8	13.0	
Clear-Sky Solar Irradiance	taub	0.227	0.248	0.263	0.316	0.322	0.332	0.359	0.347	0.302	0.266	0.244	0.235		
	taud	2.394	2.310	2.363	2.355	2.453	2.422	2.398	2.427	2.562	2.641	2.558	2.420		
	Ebn at Noon	305	311	317	304	303	298	289	290	300	304	298	292		
	Edh at Noon	25	32	35	38	35	36	37	35	28	23	22	23		
All-Sky Solar Radiation	RadAvg	707	977	1424	1746	1978	2378	2210	1896	1623	1141	770	586		
	RadStd	42	71	123	129	145	145	95	89	106	99	83	50		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page